Math Bootcamp

Day 2 — IFT, MVT, Integration, and Taylor

Math Camp 2026

2.1 Explicit vs. implicit relationships

Explicit: one variable is written directly as a function of others.

$$y = g(x) \qquad \text{e.g.} \qquad p^* = \frac{\alpha - c}{b + d}.$$

Implicit: variables are linked by an equation $F(x, y) = 0$.

$$F(x, y) = 0 \qquad \text{e.g.} \qquad E(p, \alpha) = 0 \quad \text{(market clearing)}.$$

Often we **know** F but cannot solve for $y = g(x)$ in closed form. The IFT says: under regularity conditions, y is still locally a function of x, and we can compute $\frac{dy}{dx}$ without solving explicitly.

2.1 What problem does the IFT solve?

Comparative statics question. How does an endogenous variable respond when an exogenous parameter shifts?

- Equilibrium price $p^*(\alpha)$ when demand depends on shifter α
- Optimal choice $x^*(w)$ from a first-order condition $F(x, w) = 0$
- Equilibrium objects defined by $F(x, \theta) = 0$ without closed-form solutions

Pattern: an equilibrium condition $F(x, \theta) = 0$ defines $y^*(\theta)$ implicitly. We want $\dfrac{dy^*}{d\theta}$ — the IFT delivers this from partials of F.

2.1 One-variable IFT — statement and recipe

Let $F(x, y)$ be C^1 near (x_0, y_0) with $F(x_0, y_0) = 0$. If $F_y(x_0, y_0) \neq 0$, then locally there is a unique g such that

$$y = g(x), \qquad F(x, g(x)) = 0, \qquad g(x_0) = y_0,$$

and

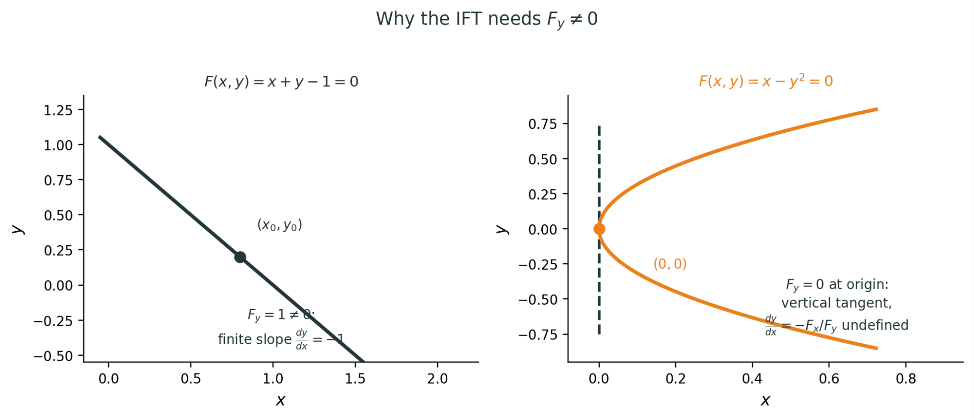
$$\boxed{g'(x) = \frac{dy}{dx} = -\frac{F_x(x, y)}{F_y(x, y)}} \qquad \text{on the curve } F(x, y) = 0.$$

Recipe. Totally differentiate $F(x, y) = 0$:

$$0 = dF = F_x\,dx + F_y\,dy \quad \Rightarrow \quad \frac{dy}{dx} = -\frac{F_x}{F_y} \quad \text{when } F_y \neq 0.$$

2.1 Why $F_y \neq 0$? — regularity condition

Left: $F_y \neq 0$ at (x_0, y_0) — tangent has finite slope; locally $y = g(x)$. **Right:** $F_y = 0$ at $(0, 0)$ — **vertical tangent**; $\dfrac{dy}{dx} = -F_x/F_y$ is undefined.



2.1 Further study: several variables — for interest

Not required for Math Camp.

System: $\mathbf{F}(\mathbf{y}, \boldsymbol{\theta}) = \mathbf{0}$, $\mathbf{F}\colon \mathbb{R}^{n+m} \to \mathbb{R}^n$.

If the $n \times n$ Jacobian $\dfrac{\partial \mathbf{F}}{\partial \mathbf{y}}$ is invertible at the point, then locally $\mathbf{y} = \mathbf{y}^*(\boldsymbol{\theta})$ and

$$\frac{\partial \mathbf{y}^*}{\partial \boldsymbol{\theta}} = -\left(\frac{\partial \mathbf{F}}{\partial \mathbf{y}}\right)^{-1} \frac{\partial \mathbf{F}}{\partial \boldsymbol{\theta}}.$$

Same pattern as the one-variable formulas with F_y and F_θ. Appears in graduate micro; camp focus is the scalar case above.

2.1 Application — setup

Market clearing. Quantity demanded equals quantity supplied:

$$D(p, \alpha) = S(p),$$

where p is price and $\alpha > 0$ is a **demand shifter** (e.g. consumer income, taste index, population).

Define excess demand

$$E(p, \alpha) = D(p, \alpha) - S(p).$$

An equilibrium price $p^*(\alpha)$ satisfies

$$E(p^*(\alpha), \alpha) = 0.$$

Question: when α rises, does equilibrium price p^* rise or fall? Use $\dfrac{dp^*}{d\alpha} = -\dfrac{E_\alpha}{E_p}$.

2.1 Application — linear example

Linear demand and supply:

$$D(p, \alpha) = \alpha - bp, \qquad S(p) = c + dp, \qquad b, c, d > 0.$$

Excess demand $E(p, \alpha) = D(p, \alpha) - S(p) = \alpha - c - (b + d)p$. Equilibrium $p^*(\alpha)$ satisfies $E(p^*(\alpha), \alpha) = 0$.

Partials: $E_p = -(b + d) \neq 0$, $E_\alpha = 1$.

IFT:

$$\frac{dp^*}{d\alpha} = -\frac{E_\alpha}{E_p} = \frac{1}{b + d} > 0.$$

2.1 Application — check by explicit solution

From $E(p^*(\alpha), \alpha) = 0$:

$$p^*(\alpha) = \frac{\alpha - c}{b + d}.$$

Differentiate explicitly:

$$\frac{dp^*}{d\alpha} = \frac{1}{b + d},$$

matching the IFT formula $-\frac{E_\alpha}{E_p}$.

When IFT is essential: nonlinear $D(p, \alpha)$ or $S(p)$ — $p^*(\alpha)$ may have no closed form, but

$$\frac{dp^*}{d\alpha} = -\frac{E_\alpha}{E_p} \qquad \text{at } p = p^*(\alpha)$$

still gives the local comparative static.

2.1 Application — economic interpretation

Chain rule. Since $E(p^*(\alpha), \alpha) = 0$ for all α,

$$0 = \frac{d}{d\alpha} E(p^*(\alpha), \alpha) = \underbrace{E_\alpha(p^*, \alpha)}_{\text{direct}} + \underbrace{E_p(p^*, \alpha)\,\frac{dp^*}{d\alpha}}_{\text{indirect via } p^*}.$$

So $\dfrac{dp^*}{d\alpha} = -\dfrac{E_\alpha}{E_p}$ when $E_p \neq 0$.

Two effects of higher α on excess demand (in words).

1. **Direct effect:** holding p fixed, higher α raises demand $\Rightarrow$ E goes **up** ($E_\alpha > 0$).
2. **Indirect effect:** equilibrium price adjusts; higher p makes E **lower** ($E_p < 0$).

Why must p^* rise? The direct effect alone would create excess demand at the old price. Price rises until the indirect effect offsets it and restores $E = 0$. With $E_\alpha > 0$ and $E_p < 0$, this requires $\dfrac{dp^*}{d\alpha} > 0$.

2.1 — Comparative statics practice (no closed form) Together

Nonlinear demand: $D(p, \alpha) = \alpha e^{-p}$, $S(p) = p$, $\alpha > 0$.

No elementary closed form for $p^*(\alpha)$.

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5. $\alpha = 2$: $2e^{-p} = p$ gives $p^* \approx 0.85$, so $\dfrac{dp^*}{d\alpha} \approx 0.23$.

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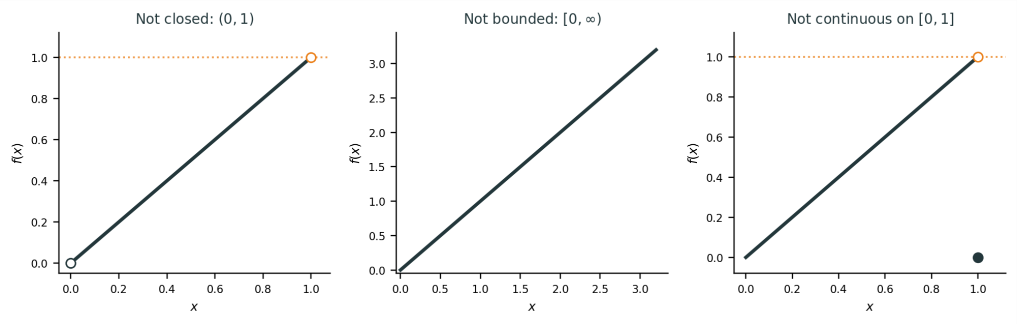
No closed form for $p^*(\alpha)$ — IFT still works.

Extreme Value Theorem (EVT). If f is **continuous** on the closed interval $[a, b]$, then f attains a **maximum** and a **minimum** on $[a, b]$.

So $\exists\, c, d \in [a, b]$ such that

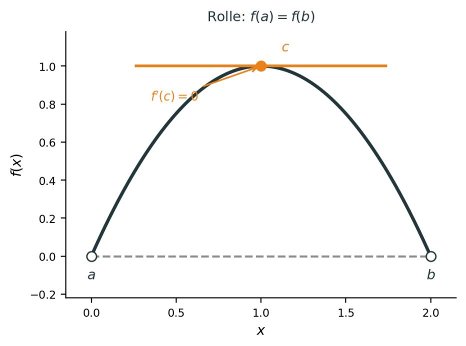
$$f(c) \leq f(x) \leq f(d) \qquad \text{for all } x \in [a, b].$$



Rolle's theorem. If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then $\exists\, c \in (a, b)$ with $f'(c) = 0$.

Proof. If f is constant on $[a, b]$, any $c \in (a, b)$ works. Otherwise, by EVT, f has a maximum or minimum on $[a, b]$. Since $f(a) = f(b)$, this extremum occurs at some **interior** point $c \in (a, b)$. At an interior peak or trough, the tangent is horizontal, so $f'(c) = 0$. $\square$

Special case of the MVT below with $f(a) = f(b)$.



2.2 Mean Value Theorem

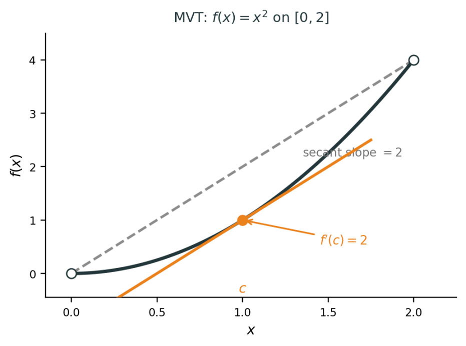
Mean Value Theorem (MVT). If f is continuous on $[a, b]$ and differentiable on (a, b), then $\exists\, c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Picture: some interior tangent is **parallel** to the secant through $(a, f(a))$ and $(b, f(b))$.

Example. $f(x) = x^2$ on $[0, 2]$: secant slope $= 2$, so $f'(c) = 2c = 2$ gives $c = 1$.

Proof idea: define $g(x) = f(x) - m(x - a)$ with $m = \dfrac{f(b) - f(a)}{b - a}$; then $g(a) = g(b)$, Rolle gives $f'(c) = m$.



2.3 Integral as area and accumulation

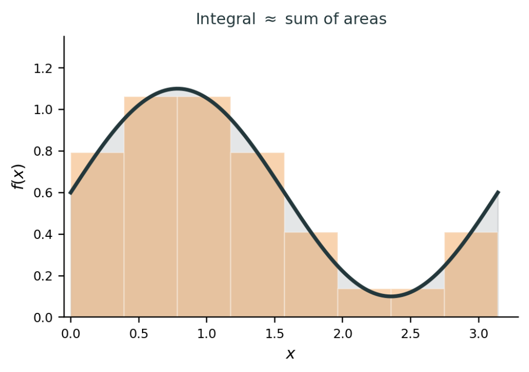
Riemann idea. For $f \geq 0$ on $[a, b]$,

$$\int_a^b f(x)\,dx \approx \sum_{i=1}^{n} f(x_i^*)\,\Delta x$$

is the **signed area** under the curve / total accumulation.

Definite vs indefinite.

- Definite: $\int_a^b f(x)\,dx$ is a number
- Indefinite: $\int f(x)\,dx = F(x) + C$ is a family of antiderivatives



Econ: total cost, consumer surplus, probability mass.

2.3 Riemann vs. Lebesgue — for interest

Not required for Math Camp.

Two ways to define $\int f$.

Riemann (this camp)	**Lebesgue (grad / probability)**
Chop the x-**axis** into small pieces	Chop the y-**axis** into levels
Sum (height) $\times$ (width)	Sum (level) $\times$ (measure of set where f hits that level)
Good for hand calculations	Integrates more pathological functions
Area under curve picture	Foundation for modern probability

Contrast in one sentence. Riemann slices **vertically**; Lebesgue slices **horizontally** by value.

2.3 Lebesgue integral — for interest

Not required for Math Camp.

When Riemann fails. Riemann needs the function to be "well behaved" on intervals: too many jumps can prevent upper and lower sums from converging to the same number.

Classic example (not Riemann integrable). Dirichlet function on $[0, 1]$:

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Every interval contains rationals and irrationals, so upper and lower Riemann sums never agree — no Riemann integral.

Lebesgue handles this. It integrates by **level sets**, so a "sparse" jump set like $\mathbb{Q}$ does not block integration the way Riemann's interval partitions do.

2.3 Fundamental Theorem of Calculus

Part I. If $F'(x) = f(x)$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Part II. If f is continuous and $F(x) = \int_a^x f(t)\,dt$, then $F'(x) = f(x)$.

Differentiation and integration are inverse operations (MVT underlies many proofs).

2.3 Integration by parts

From the product rule. For differentiable $u(x)$ and $v(x)$,

$$(uv)' = u'v + uv' \quad \Rightarrow \quad uv' = (uv)' - u'v.$$

Integrate both sides:

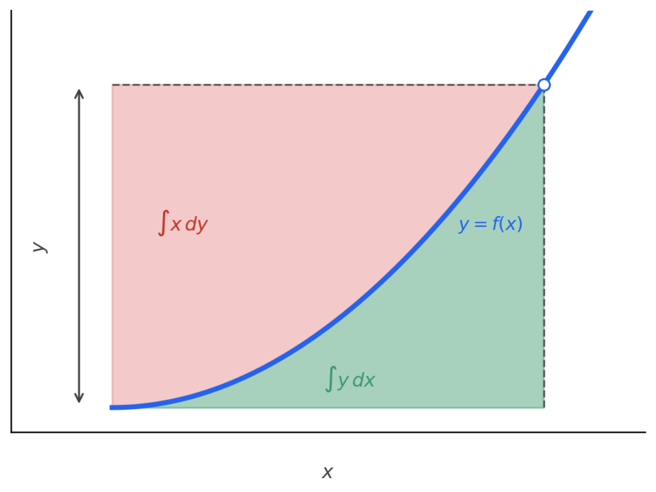
$$\int u\,v'\,dx = uv - \int u'v\,dx \qquad \text{or} \qquad \boxed{\int u\,dv = uv - \int v\,du}.$$

On $[a, b]$ with $u'v$ integrable,

$$\int_a^b u(x)v'(x)\,dx = \Big[u(x)v(x)\Big]_a^b - \int_a^b u'(x)v(x)\,dx.$$

Pick u and dv so the new integral $\displaystyle\int v\,du$ is simpler.

2.3 Integration by parts (graph)



2.3 Integration by parts — example

Example. $\int xe^x\,dx$.

Let $u = x$, $dv = e^x\,dx$, so $du = dx$, $v = e^x$:

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C = e^x(x-1) + C.$$

Check. $\frac{d}{dx}\left[e^x(x-1)\right] = e^x(x-1) + e^x = xe^x$. ✓

2.3 PMF and CMF — discrete example (dice)

Setup. Fair die: $X \in \{1, 2, 3, 4, 5, 6\}$.

Probability mass function (PMF): $p(k) = P(X = k) = \frac{1}{6}$ for $k = 1, \ldots, 6$.

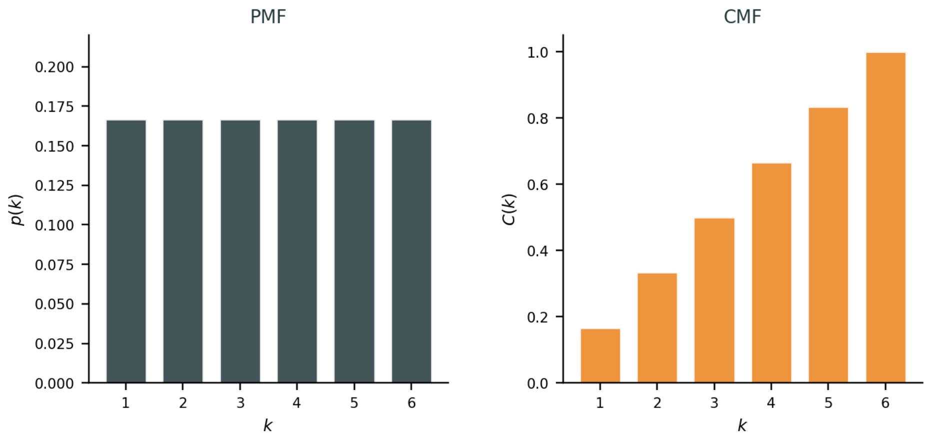
Event (sum rule): $P(2 \leq X \leq 4) = p(2) + p(3) + p(4) = \frac{1}{2}$.

Cumulative mass function (CMF): $C(k) = P(X \leq k) = \sum_{j \leq k} p(j)$, defined only at the **outcomes** $k = 1, \ldots, 6$ (not a continuous curve in x).

Example: $C(3) = p(1) + p(2) + p(3) = \frac{1}{2}$.

Discrete: **PMF/CMF** and **sums**. Continuous (next slides): **PDF/CDF** and **integrals**.

2.3 PMF and CMF — dice (graph)



2.3 PDF and CDF — continuous case

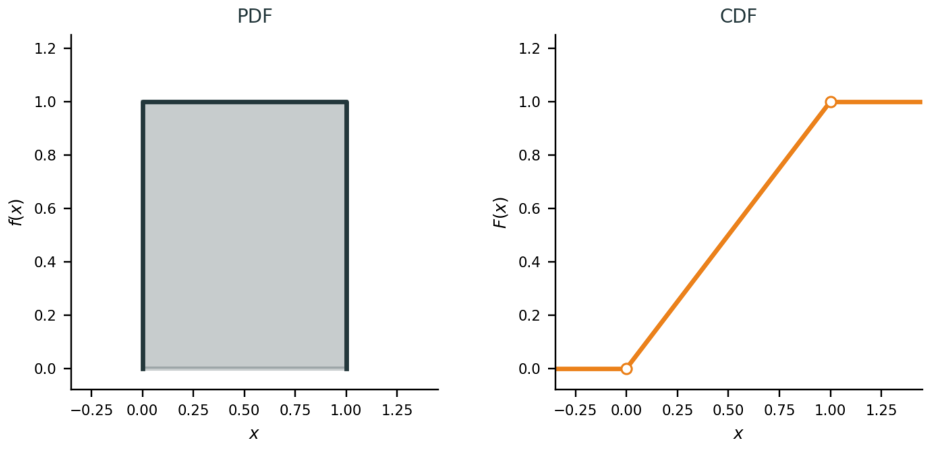
Let X be a continuous random variable.

Probability density function (PDF): $f(x) \geq 0$ and $\int_{-\infty}^{\infty} f(x)\,dx = 1$. Then $P(a \leq X \leq b) = \int_a^b f(x)\,dx$.

Cumulative distribution function (CDF): $F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt$; a **continuous** curve in x (not jumps on a finite outcome list).

FTC: if f is continuous, then $F'(x) = f(x)$.

2.3 PDF and CDF — Unif$(0, 1)$ (graph)



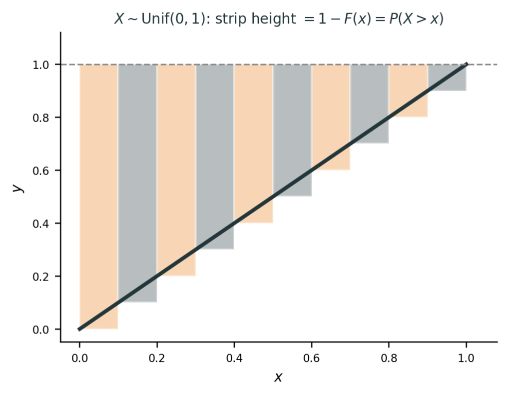
2.3 Expectation from the CDF

For a **non-negative** continuous X with PDF f and CDF F,

$$\mathbb{E}[X] = \int_0^\infty xf(x)\,dx = \int_0^\infty (1-F(x))\,dx = \int_0^\infty P($$

Intuition. For $X \sim \mathrm{Unif}(0,1)$, each vertical strip has height $1 - F(x) = P(X > x)$; $\mathbb{E}[X]$ is their total area.

Derivation via integration by parts: appendix.



2.4 Leibniz rule — fixed limits

Setup. $F(\theta) = \displaystyle\int_a^b f(x,\theta)\,dx$ with fixed a, b.

Leibniz rule (fixed limits). When regularity holds,

$$\frac{dF}{d\theta} = \int_a^b \frac{\partial f}{\partial \theta}(x,\theta)\,dx.$$

Regularity (enough for this camp).

- $\dfrac{\partial f}{\partial \theta}(x,\theta)$ exists
- f and $\dfrac{\partial f}{\partial \theta}$ are **continuous** in (x,θ)

Then you may swap $\dfrac{d}{d\theta}$ and $\displaystyle\int_a^b$.

Idea. Only the integrand moves with θ; the limits a, b stay fixed.

When regularity fails (moving jump): appendix.

2.4 Leibniz rule — variable limits

Setup. $F(\theta) = \displaystyle\int_{a(\theta)}^{b(\theta)} f(x,\theta)\, dx$.

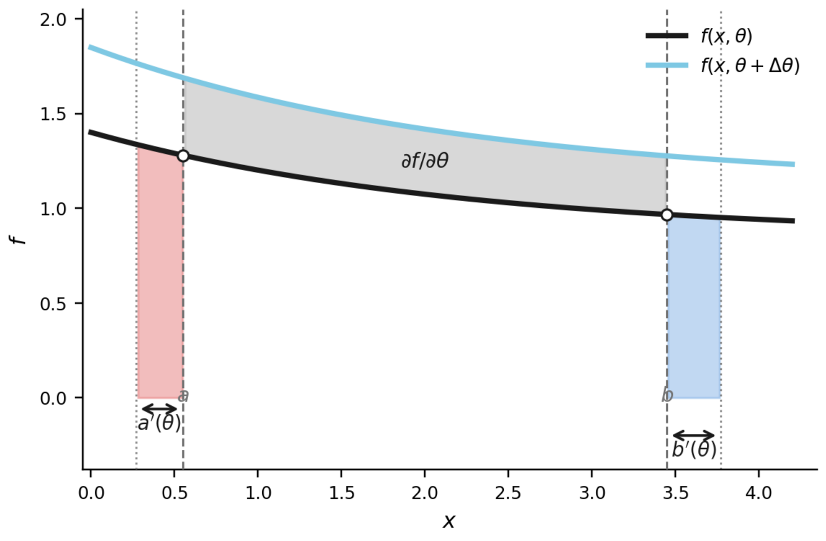
Leibniz rule (variable limits). When $a(\theta)$, $b(\theta)$ are differentiable and regularity holds,

$$\frac{dF}{d\theta} = f\big(b(\theta),\theta\big)\, b'(\theta) - f\big(a(\theta),\theta\big)\, a'(\theta) + \int_{a(\theta)}^{b(\theta)} \frac{\partial f}{\partial \theta}(x,\theta)\, dx.$$

Three parts. (i) area added/lost at the **upper** limit b; **(ii)** at the **lower** limit a; **(iii)** change **inside** the interval $(\partial f/\partial\theta)$.

Graph on next slide: grey = $\partial f/\partial\theta$; pink/blue = lower/upper limit terms ($a'(\theta)$, $b'(\theta)$).

2.4 Leibniz rule — variable limits (graph)



2.4 Fubini's theorem

Statement. If f is **integrable** on region D, i.e.

$$\iint_D |f(x,y)|\,dx\,dy < \infty,$$

then

$$\iint_D f(x,y)\,dx\,dy = \int\left(\int f(x,y)\,dx\right)dy = \int\left(\int f(x,y)\,dy\right)dx,$$

(inner limits depend on the shape of D).

Camp shortcut. If D is bounded and f is **continuous** on D, then f is integrable and Fubini applies.

Tonelli (for interest): if $f \geq 0$ and measurable, the same swap holds without checking $\iint |f|$ first.

2.4 Fubini — triangle example

Example (triangle).

$$D = \{(x, y) : 0 \le x \le 1,\ 0 \le y \le x\}.$$

Compute $\displaystyle\iint_D (x + y)\, dx\, dy$.

y **first, then** x**:** for fixed x, $y \in [0, x]$:

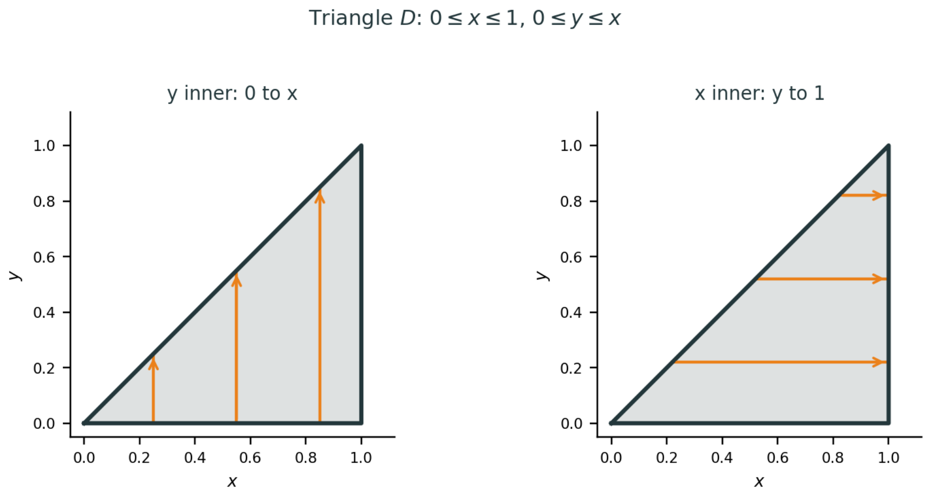
$$\int_0^1 \int_0^x (x + y)\, dy\, dx = \int_0^1 \left[xy + \tfrac{y^2}{2}\right]_0^x dx = \int_0^1 \tfrac{3x^2}{2}\, dx = \tfrac{1}{2}.$$

x **first, then** y**:** for fixed y, $x \in [y, 1]$:

$$\int_0^1 \int_y^1 (x + y)\, dx\, dy = \int_0^1 \left[\tfrac{x^2}{2} + xy\right]_y^1 dy = \int_0^1 \left(\tfrac{1}{2} + y - \tfrac{3y^2}{2}\right) dy = \tfrac{1}{2}.$$

Graph on next slide: same triangle, different slice directions.

2.4 Fubini — triangle (graph)



2.5 First-order Taylor and linearization

One variable. Near a,

$$f(x) \approx f(a) + f'(a)(x-a).$$

Several variables. Near $\mathbf{a}$,

$$f(\mathbf{a}+\mathbf{h}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} \qquad \text{(Day 1 tangent plane).}$$

Comparative statics reading. A small parameter shift $\Delta\theta$ changes an equilibrium object by

$$\Delta x^* \approx \frac{dx^*}{d\theta}\,\Delta\theta,$$

i.e. **local linearization** — first-order Taylor.

Second order (preview): $f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2$ captures curvature; needed when linearization is poor.

2.5 When linearization works

Works well when:

- the shift is **small**
- curvature is modest (f'' not huge)
- the function is differentiable (tangent plane exists)

Fails or misleads when:

- large shifts, kinks, or non-differentiability
- strong nonlinearity (e.g. corner solutions, thresholds)

Link to Day 2 tools. IFT gives exact first-order comparative statics; Taylor explains why that linear approximation is the natural local forecast.

A.1 Expectation from the CDF

Claim. For non-negative continuous X with PDF f and CDF F,

$$\mathbb{E}[X] = \int_0^\infty x f(x)\, dx = \int_0^\infty \big(1 - F(x)\big)\, dx.$$

Proof (integration by parts). Set $u = 1 - F(x)$ and $dv = dx$. Then $du = -f(x)\, dx$ and $v = x$:

$$\int_0^\infty (1 - F(x))\, dx = \big[x(1 - F(x))\big]_0^\infty - \int_0^\infty x\,(-f(x))\, dx = \big[x(1 - F(x))\big]_0^\infty + \int_0^\infty x f(x)\, dx.$$

At $x = 0$, $x(1 - F(x)) = 0$. As $x \to \infty$, $x(1 - F(x)) \to 0$ when $\mathbb{E}[X] < \infty$. Hence $\int_0^\infty (1 - F(x))\, dx = \mathbb{E}[X]$. $\square$

A.2 Leibniz rule — failure example

Example (fixed limits). On $[0,1]$, let

$$f(x,\theta) = \mathbf{1}\{x > \theta\} \qquad \text{(step jumps at } x = \theta\text{)}.$$

Then

$$F(\theta) = \int_0^1 f(x,\theta)\,dx = 1 - \theta \quad \Rightarrow \quad F'(\theta) = -1.$$

Why Leibniz fails. The discontinuity **moves with** θ, so $\partial f/\partial\theta$ is not continuous on $[0,1] \times I$. Naively $\partial f/\partial\theta = 0$ a.e., so $\int_0^1 0\,dx = 0 \neq F'(\theta)$.

A.2 Leibniz failure — moving step (graph)